Zero-Shot Forensic Face Synthesis via Text-Conditioned Diffusion

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Fig. 1: Each unit displays the input sketch (left), real photo (center), and generated output (right).

***Abstract*—Cross-modal forensic sketch-to-photo matching remains challenging, as contemporary facial recognition systems struggle to bridge the domain gap between sparse line drawings and high-fidelity photographs. Furthermore, standard Generative AI frameworks frequently prioritize aesthetic quality over identity preservation, yielding synthetic outputs that lack the biometric fidelity required for forensic identification. This paper proposes an AI-assisted pipeline based on the Stable Diffusion v1.5 architecture designed to mitigate these shortcomings. Our system integrates a line-art conditioned ControlNet using RealisticVision to enforce strict geometric alignment with the source sketch, alongside a specialized LoRA (Low-Rank Adaptation) module to refine the diffusion process toward photorealistic facial synthesis. By implementing architectural modifications to the standard ControlNet pipeline, we enhance the model's ability to maintain structural identity while generating forensic-grade imagery. Experimental results demonstrate that our approach significantly reduces identity drift compared to other approaches, providing a more reliable tool for forensic template matching. Code and models are publicly available at: https://github.com/rachit6105/Sketch-2-Image**

***Index Terms*—forensic AI, sketch to photo,stable diffusion, controlnet, longclip, lora**

I. Introduction

In recent years, bridging the domain gap between sparse, hand-drawn sketches and realistic photographs has become critical for forensic applications. Traditional facial recognition systems rely heavily on colored, textured photographs, making it incredibly difficult to identify suspects directly from forensic sketches. To tackle this, the research community has explored various image-to-image translation techniques, leveraging popular architectures like Generative Adversarial Networks (GANs) and diffusion models. With the rapid advancement of generative AI, these models demonstrate impressive capabilities in synthesizing highly detailed images from text or structural prompts, offering a promising technological solution to this cross-modal challenge.

However, applying these foundational models to human faces introduces a severe limitation for forensic utility: an inherent "beautification bias." In forensic science, preserving exact, unretouched facial structures is far more important than creating an aesthetically pleasing image. The root of this problem lies in the underlying training data. Because most popular models are trained either on broad,

general image datasets [1], [2], [3] or on repositories heavily skewed toward celebrity faces, such as CelebA [4] and FairFace [5], they are inherently biased toward generating symmetrical, glossy, and highly idealized facial priors. Consequently, during synthesis, they naturally smooth out the raw, asymmetrical phenotypic imperfections that are absolutely vital for accurate biometric identification. In doing so, the true identity of the subject is often altered or completely lost. Furthermore, because diverse racial features are frequently underrepresented in these datasets, the models struggle to accurately reproduce a wide spectrum of demographic characteristics, such as Asian and African facial traits. Overcoming this generative bias to preserve true forensic fidelity is the primary motivation of this work.

Furthermore, existing solutions predominantly suffer from demographic overfitting, lacking true **zero-shot transferability** across a diverse spectrum of racial and ethnic identities. Additionally, current frameworks rely almost exclusively on the spatial constraints of the sketch, systematically failing to utilize the rich, qualitative textual descriptions provided by eyewitnesses—a modality that is crucial for resolving ambiguous facial traits. To address these critical limitations, we propose a multimodal conditional diffusion pipeline. By combining dense textual conditioning with the geometric enforcement of a ControlNet[6] framework, our approach directly translates eyewitness descriptions into localized feature guidance. Through the strategic integration of specialized pipeline components—including extended textual encoders and targeted domain adaptation—our system bridges the modality gap, achieving robust, zero-shot demographic transferability without requiring a ground-up architectural redesign.

II. Related Work

A. Generative Adversarial Networks and Dataset Bias

Previous research has extensively explored image-to-image translation to synthesize photographs from sketches. Architectures leveraging Generative Adversarial Networks (GANs) [7] or StyleGAN [8][9] as the base architecture can produce visually striking results but suffer from severe domain rigidity.Further approaches like Denoising Diffusion Probabilistic Models (DDPMs) [10] which uses a CycleGAN as Coarse generator for developing a vaguely defined coloured image as a preprocessing step.These models generally succeed only when the input sketch strictly aligns with the spatial distribution of their specific training dataset. For example, models trained exclusively on constrained academic datasets like the CUFS[11] frequently overfit to the dominant demographic of that data. Consequently, they act as rigid templates, defaulting to generating subjects with specific characteristics (e.g., fair skin and straight black hair) regardless of the input sketch geometry, completely failing to achieve zero-shot transferability across diverse racial and ethnic profiles.

B. The Synthetic Sketch Fallacy and Hallucination

A critical flaw in a significant portion of recent sketch-to-photo literature is the reliance on synthetic data for methodology evaluation. Many contemporary frameworks like those using GANs[9] utilize high-resolution photographic datasets and algorithmically extract Canny edge maps or synthetic line-art to serve as proxy "sketches". There are models like DeepFace[12] and DiffFaceSketch[13] which utilise face outline doodles instead of detailed sketch. While models trained on this synthetic pipeline perform fine on perfect edge maps, they suffer severe degradation when exposed to actual, hand-drawn forensic sketches. Moreover these models are more suited to general face generation and cannot be trusted for identity retention.They tend to hallucinate on hand drawn sketches and often form blurry and smudged outputs.

C. Identity Drift in Unconstrained Diffusion

More recently, standard and large-scale text-to-image foundational models have been applied to facial synthesis. While they offer superior stability and higher resolution over GANs, they introduce a distinct, highly detrimental issue: extreme identity drift. Because these models lack strict spatial conditioning, they tend to generate a structurally generic face that merely resembles the input, rather than preserving the specific biometric identity of the subject. Furthermore, the iterative denoising process inherently suppresses high-frequency spatial anomalies. This actively "beautifies" the output, resulting in unnaturally smooth, plastic-like skin textures that often induce an uncanny valley effect. This loss of unretouched, asymmetric imperfections renders the generated photos forensically invalid.

III. Background

A. Conditional Diffusion Models

To bridge the gap from purely text-to-photo synthesis to structurally guided image-to-photo generation, additional conditioning mechanisms are required. ControlNet [6] extends pre-trained diffusion models by freezing the base model and adding trainable neural network blocks that condition generation on structured spatial inputs, such as depth maps, human poses, or edge maps.In this project, we employ a line-art conditioned ControlNet to preserve the precise geometric structure of the input sketch.

B. Generating Photos via Diffusion

Diffusion models, particularly Latent Diffusion Models (LDMs) [**?**], have emerged as the state-of-the-art architecture for high-fidelity image synthesis. These models operate by iteratively denoising a normally distributed variable to recover a structured image latent space. In this work, we utilize the Stable Diffusion (SD v1.5)[14] architecture as our foundational base model due to its robust prior knowledge of facial structures and its broad compatibility with auxiliary conditioning modules.

To guide this generative denoising process, text-to-image models heavily rely on Contrastive Language-Image Pre-training (CLIP) [15] encoders. The standard CLIP text encoder translates user prompts into semantic embeddings that condition the denoising UNet via cross-attention mechanisms. We integrate LongCLIP [16], a framework that extends the context window and enhances fine-grained detail retention, allowing the model to process comprehensive forensic descriptions effectively.

C. Latent Space Compression and VAEs

Unlike standard pixel-space diffusion models, Stable Diffusion is a Latent Diffusion Model (LDM). To achieve computational efficiency, LDMs do not process high-resolution images directly. Instead, they rely on a pre-trained Variational Autoencoder (VAE) to compress images into a lower-dimensional latent space before the diffusion process and subsequently decode the generated latents back into pixel space. The choice of VAE guides how the resulting image looks, even for the same latent space vectors.

D. Parameter-Efficient Fine-Tuning

Adapting foundational diffusion models to specialized domains conventionally requires computationally expensive full-parameter fine-tuning. Low-Rank Adaptation (LoRA) [17] offers a parameter-efficient alternative by injecting small, trainable low-rank matrices into the attention layers of the frozen base model.

E. Face Matching and Evaluation

Evaluating the efficacy of forensic sketch-to-photo pipelines requires robust, identity-preserving metrics rather than pure aesthetic assessments. To quantitatively measure biometric fidelity, modern face recognition paradigms rely on deep feature embeddings. Models like ArcFace[18] or face_recognition library [19], which is built upon the deep learning algorithms provided by the dlib toolkit [20] that maps facial images into a highly discriminative hyperspherical embedding space, maximizing intra-class similarity while penalizing inter-class variance.To establish a definitive decision boundary for whether a generated facial photo successfully preserves the original identity, we map the facial images into a low-dimensional embedding space using a pre-trained facial recognition network. By measuring the spatial and angular distances between these latent vectors, we can mathematically argue the fidelity of the biometric preservation.

IV. Proposed Methodology

A. Description Generation and Textual Conditioning

To bridge the modality gap between a hand-drawn sketch and a photorealistic facial image, our pipeline incorporates dense textual descriptions of facial attributes to guide the generative process.

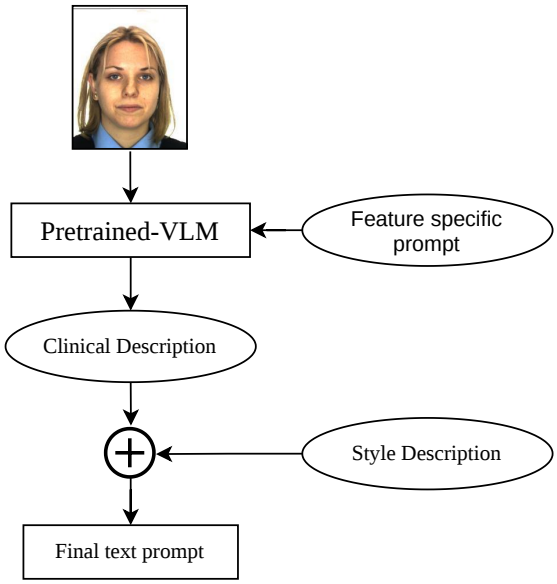

Fig. 2: Automated description generation for training.

1) Automated Clinical Description Generation via Vision-Language Model: During dataset preparation, we use a vision-language model accessed through the Google Generative AI API to analyze the ground-truth reference photographs and generate compact yet detailed facial descriptions. The goal is to capture facial attributes that are useful for forensic-style image synthesis, including age cues, face shape, jawline, hairline, hair state, eye shape, eyebrows, nose, lips, skin texture, facial hair, visible scars, moles, and eyewear.

The model is prompted with the following instruction:

```
Act as an observant witness describing
someone to a police sketch artist. Describe
the person in a single, cohesive paragraph.
CRITICAL WORD LIMIT: The description MUST
be strictly under 150 words. Be extremely
concise. Use dense, comma-separated keywords
and short phrases instead of long flowing
sentences. Focus entirely on: estimated
age, gender, race/ethnicity, skin tone,
face geometry, jawline. Hair: Describe
the hairline, color, texture, and EXACT
STATE (e.g., tied up, ponytail, bun, loose,
bald). Eyes/Face: Describe eyebrows, eye
shape/color, nose, lips, and specific skin
texture (pores, wrinkles). CRITICAL: You
MUST explicitly state if the person is
wearing glasses, spectacles, or any eyewear.
Explicitly list any moles, scars, or facial
hair. CRITICAL FORMATTING: Output ONLY the
physical description. No intro or outro
text. Do not say "Here is the description".
Start the very first word with the physical
description.
```

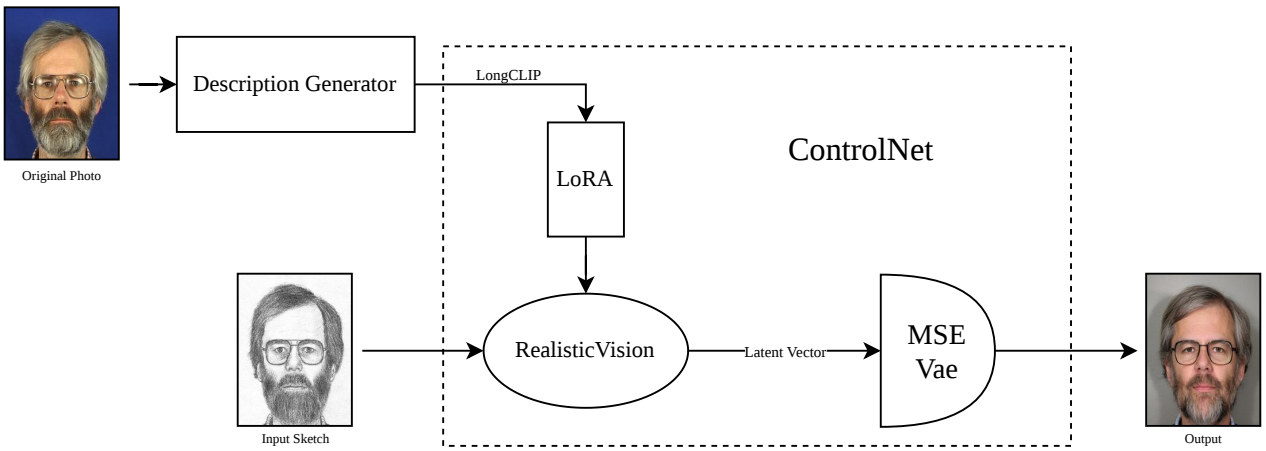

Fig. 3: Illustration of the proposed method. The description generator produces a detailed facial description, which is then concatenated with a style-based prompt.

This produces a structured witness-style facial description for each training image, which is stored and later used as a conditioning signal during generation.

2) Style Prompt Concatenation: To encourage a forensic and less beautified visual appearance, the generated facial description is concatenated with a fixed style-oriented prompt during inference. This base prompt is designed to steer the model toward raw, documentary-style imagery with visible skin detail, harsh illumination, and minimal aesthetic enhancement. The exact style prompt used in our implementation is:

```
Raw amateur photo, DMV camera mugshot,
(extreme skin texture:1.4), (visible
pores:1.3),hyper-detailed unairbrushed
skin, harsh flash lighting, documentary
photography.
```

Thus, the final positive conditioning text is formed as the concatenation of the generated clinical facial description and the above style prompt. In addition, we employ a separate negative prompt to explicitly suppress undesired generative priors such as beautification, stylization, synthetic rendering artifacts, and overly idealized facial structure. The negative prompt used is:

```
deformed iris, deformed pupils,
semi-realistic, cgi, 3d, render, sketch,
cartoon, drawing, anime:1.4,text, close
up, cropped, out of frame, worst quality,
jpeg artifacts,airbrushed, retouched,
makeup, foundation, instagram, glamour shot,
ultra-HD, 4K,studio lighting, soft focus,
plastic, doll, 3d render, beauty filter
```

This negative conditioning discourages the model from producing cartoon-like, glamour-enhanced, airbrushed, or artificially symmetric outputs, thereby preserving the intended raw and realistic forensic aesthetic. During generation, the positive prompt, negative prompt, and sketch-conditioned ControlNet input are jointly provided to the diffusion model.

3) Extended Token Processing via LongCLIP: Because the final conditioning text is formed by concatenating a dense facial description with a style-enforcing prompt, the resulting sequence can exceed the 77-token limit of the standard CLIP text encoder used in Stable Diffusion. Such truncation would discard potentially important facial details.

To address this, we employ LongCLIP-L-Diffusers[1] as the text encoder. LongCLIP extends the effective context window to 248 tokens, allowing the model to preserve substantially more of the combined prompt. This helps retain both the global forensic style cues and the finer-grained facial attributes in the text embedding used for image generation.

B. Geometric Constraint via Line-Art ControlNet

While textual descriptions dictate the semantic and textural properties of the generated image, they cannot enforce exact spatial geometry. To ensure the generated photograph aligns perfectly with the forensic artist's structural intent (e.g., jawline angle, interpupillary distance), we employ a line-art conditioned ControlNet. The ControlNet branch `control_v11p_sd15_lineart`[2] is used to preserve the geometry of the sketch. Since witness sketches provide the overall face shape, placement of eyes, jaw contour, and hairstyle silhouette, this spatial guidance is essential for sketch-to-image alignment.During preprocessing, the input sketch is inverted to conform to the expected input distribution of the ControlNet module. The ControlNet extracts the geometric features and injects them directly into the UNet's downsampling and middle blocks, locking the spatial layout of the facial features across all denoising steps. The sketch image is:

- resized to 512×512,

[1]https://huggingface.co/zer0int/LongCLIP-L-Diffusers
[2]https://huggingface.co/lllyasviel/control_v11p_sd15_lineart

- converted to RGB,
- inverted before being passed to ControlNet.

The inversion step was included because, in practice, it better aligned the sketch input with the conditioning behavior of the chosen lineart pipeline configuration.

C. Base Prior Selection and Latent Decoding

To minimize the distance between the model's base distribution and the target forensic domain, we bypassed the standard SD 1.5 weights and *Realistic Vision V5.1*[3]. Compared to plain SD v1.5 variants, it showed better rendering of realistic skin, reduced synthetic smoothness, and more natural facial texture. Since the model is provided without an integrated VAE, a separate VAE is injected.

Furthermore, because Stable Diffusion operates in a compressed latent space, the final decoding step is highly susceptible to losing high-frequency details. We observed that default Variational Autoencoders (VAEs) frequently smoothed out critical micro-textures such as pores and harsh lighting artifacts. Consequently, we explicitly injected an MSE-fine-tuned VAE `sd-vae-ft-mse`[4] into our pipeline. This specialized decoder ensures that the raw, documentarian textures generated in the latent space are faithfully reconstructed in the final pixel-space output.

D. Domain Adaptation and Bias Suppression

The final and most critical component of our methodology addresses the pervasive beautification bias. Even with a realistic base model, diffusion architectures naturally gravitate toward idealized, symmetrical outputs. We counteract this through two simultaneous mechanisms:

1) Forensic LoRA Adaptation: We trained and applied a custom Low-Rank Adaptation (LoRA) module specifically targeted at raw, unretouched facial phenotypes. By injecting these low-rank matrices into the cross-attention layers of the UNet, we shift the generative prior toward authentic skin textures and natural asymmetries, anchoring the output in forensic realism.

2) Aggressive Classifier-Free Guidance (CFG): To mathematically penalize idealized generations, we employed a strict Classifier-Free Guidance strategy (Guidance Scale = 5.0) paired with an extensive negative prompt. By explicitly conditioning the negative trajectory with terms such as *"airbrushed, perfect skin, symmetrical, glamour shot,"* the model is forcefully steered away from the aesthetic latent space and restricted to generating raw, unflattering, and biometrically accurate phenotypes.

E. Training Phase

During training, the ground-truth target photograph is available. To construct a highly accurate, multimodal training triplet (sketch, text, photo), we utilized a Vision-Large Language Model (Google Generative AI API). The model was prompted with strict constraints to analyze the ground-truth photo and automatically generate a dense, clinical textual description of the subject's facial features. This generated text, concatenated with standard base prompts, was fed alongside the corresponding sketch into the ControlNet and VAE pipeline to optimize the LoRA weights. This synthetic captioning process ensures the text embeddings perfectly align with the visual micro-textures of the target image.

F. Inference Phase

Conversely, during real-world inference, the ground-truth photograph is entirely absent. The system relies strictly on two inputs: the hand-drawn spatial sketch provided by the forensic artist, and the qualitative textual description provided by eyewitnesses or law enforcement. Because the LoRA and base model were conditioned on dense clinical text during training, the pipeline is able to successfully interpret these witness descriptions and bridge the modality gap, generating a high-fidelity suspect photograph.

V. Experimental Setup

Table I summarizes the core system.

TABLE I: Main components of the current pipeline

Component	Model / Tool	Purpose
Base model	`SG161222/Realistic_Vision_V5.1_noVAE`	Photorealistic generation with stronger skin realism
VAE	`stabilityai/sd-vae-ft-mse`	Improves color quality and local image fidelity
ControlNet	`lllyasviel/control_v11p_sd15_lineart`	Enforces sketch-based spatial alignment
Text encoder	`zer0int/LongCLIP-L-Diffusers`	Supports longer descriptive text inputs
LoRA Modules	Custom Forensic, Age, and Skin Tone	Mitigates aesthetic bias and enables precise biometric control
Prompting	Hand-crafted prompts	Encourages forensic, unretouched appearance
Assembly	PIL image composition	Produces comparison layout

A. Dataset Curation and Preprocessing

The primary data was sourced from the CUHK Face Sketch (CUFS) database [21], which we expanded by integrating several other benchmark repositories, including AR [22], CUFSF [23], XM2VTS [24], and the highly unconstrained WildSketch dataset [25]. This specific aggregation strategy is critical to our methodology. Standard models trained exclusively on constrained datasets (e.g., only CUFS) frequently suffer from demographic overfitting. By incorporating highly varied datasets like WildSketch, which introduces unconstrained, real-world forensic sketching styles, we ensure a diverse spatial distribution of facial geometries, varying lighting conditions, and diverse ethnic profiles.

[3]https://huggingface.co/SG161222/Realistic_Vision_V5.1_noVAE
[4]https://huggingface.co/stabilityai/sd-vae-ft-mse-original

The fully curated dataset was partitioned to ensure no data leakage between phases. The final split consists of **2,668 paired images** utilized for the training phase, and a rigorously isolated testing set of **326 images**reserved for zero-shot inference and quantitative evaluation.

To facilitate our multi-modal architecture, the ground-truth photographic pairs from the training set were individually processed through our Vision-LLM framework. This generated the dense, clinical textual descriptions required for the LongCLIP encoder, effectively transforming the traditional bipartite photo-sketch data into a rich, triplet dataset consisting of the spatial sketch, the localized clinical text, and the target photographic output.

B. Training Environment

The proposed multi-modal pipeline was implemented utilizing the PyTorch deep learning framework. To handle the complex diffusion trajectories and extended textual encodings, our architecture heavily leveraged the Hugging Face `diffusers` and `transformers` libraries.

Due to the distinct computational requirements of our two-stage pipeline, the hardware environment was explicitly partitioned between training and inference. The computationally intensive training phase—which encompassed the fine-tuning of the ControlNet spatial weights and the optimization of the domain-adaptation LoRA matrices—was executed on the PARAM SANGANAK high-performance computing cluster. Specifically, training operations were accelerated using dedicated GPU compute nodes, each equipped with dual NVIDIA Tesla V100 GPUs (16 GB HBM2 VRAM per card), supported by dual Intel Xeon G-6248 processors (40 cores) and 192 GB of DDR4 memory.

Conversely, the inference phase and all quantitative evaluations were conducted on a localized setup to demonstrate the deployment efficiency and accessibility of our trained pipeline. All zero-shot generations, sketch-to-photo translations, and biometric identity verifications were seamlessly executed on a local workstation accelerated by a single, consumer-grade NVIDIA GeForce RTX 3070 Ti GPU (8 GB VRAM).

C. Dataset Preparation

To ensure the generative pipeline produces accurate, forensic-style composites rather than idealized portraits, a custom dataset of 2,668 mugshot-style training images was curated. The data preparation process was automated to ensure consistency at scale. Structured clinical descriptions detailing specific facial attributes—such as age, ethnicity, face shape, skin texture, and visible marks—were generated using the Gemini API and compiled into a master JSON file. Crucially, to anchor the learned documentary aesthetic during training and inference, the unique trigger word `forensic_mugshot_style` was integrated into every description. A processing script then parsed this JSON, extracting these augmented descriptions and saving them as individual `.txt` files paired directly with their corresponding `.jpg` training images, ensuring a robust text-image pairing for the training phase.

D. Training Approach

The custom forensic LoRA was trained using the `sd-scripts` Kohya pipeline, which is highly optimized for Stable Diffusion fine-tuning. The primary objective was to adapt the UNet attention modules to learn the harsh, unairbrushed aesthetic of documentary mugshots while keeping the base weights of `SG161222/Realistic_Vision_V5.1_noVAE` frozen.

The training was executed on PARAM SANGANAK high-performance computing cluster. The training environment leveraged `xformers` for memory-efficient attention and was configured with the parameters detailed in Table II.

TABLE II: Kohya LoRA Training Configuration

Parameter	**Value**
Base model	`SG161222/Realistic_Vision_V5.1_noVAE`
Resolution	512×512
LoRA rank & alpha	16
Max train steps	1500
Learning rate	1×10^{-4} (Constant Scheduler)
Optimizer	AdamW
Mixed precision	fp16
CLIP skip	2
Max token length	225

E. Inference Integration

Because the Kohya architecture was used, the resulting weights (`forensic_kohya.safetensors`) are natively compatible with Diffusers. During inference, this adapter is seamlessly loaded into the pipeline alongside the pre-trained Age and Skin Tone sliders [26], [27]. Combined with strict negative prompting against idealized traits, this multi-LoRA architecture provides precise, multidimensional control over the suspect's appearance.

F. Interactive User Interface

To facilitate interactive forensic facial synthesis, a frontend interface was developed using the Gradio framework. The application employs a structured two-column layout to clearly separate user inputs from the generated results. The primary input panel allows users to upload a line-art police sketch, enter a detailed clinical description, and precisely control the generation parameters through three dynamic sliders that adjust the weights of the forensic, skin tone, and age LoRAs. Additionally, the interface features a real-time status indicator to verify active model adapters. Upon execution, the refined photorealistic composite is rendered in the adjacent output panel, enabling a seamless and iterative generation process.

VI. Testing and Results

A. Evaluation Strategy

At the current stage, evaluation is primarily qualitative. Generated images are compared with ground-truth photos using visual inspection. The side-by-side output grid supports inspection of geometry preservation, identity resemblance, realism, faithfulness to clinical details, and undesirable beautification or stylization.

To establish a definitive decision boundary for whether a generated facial photo successfully preserves the original identity, we map the facial images into a low-dimensional embedding space using a pre-trained facial recognition network; we use InsightFace. Let $\mathbf{u}$ and $\mathbf{v}$ represent the n-dimensional feature embeddings extracted from the generated output and the ground-truth reference, respectively. By measuring the spatial and angular distances between these latent vectors, we can mathematically argue the fidelity of the biometric preservation.

The first metric employed is the Euclidean Distance (L_2 norm), which calculates the absolute straight-line spatial difference between the embeddings:

$$d_E(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{\sum_{i=1}^{n} (u_i - v_i)^2} \tag{1}$$

A minimization of the Euclidean distance indicates tighter geometric clustering in the latent space, suggesting a high probability that the generated face falls within the correct identity threshold.

To complement the spatial distance, we utilize Cosine Similarity, which evaluates the angular orientation of the feature vectors independent of their magnitude. This is particularly critical in facial recognition, as it is highly robust to variations in lighting or contrast that might scale the vector but preserve its underlying biometric pattern. It is calculated as the dot product of the vectors normalized by their magnitudes:

$$S_C(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \frac{\sum_{i=1}^{n} u_i v_i}{\sqrt{\sum_{i=1}^{n} u_i^2} \sqrt{\sum_{i=1}^{n} v_i^2}} \tag{2}$$

The resulting similarity score is bounded between -1 and 1. A score approaching 1 strongly argues that the generated features align perfectly with the original subject's identity vector. Furthermore, if the extracted embeddings are strictly L_2-normalized prior to comparison, these two metrics become mathematically unified, providing a highly stable decision boundary for zero-shot identity verification.

B. Results

1) Impact of Description Conditioning: We conducted an isolation test where the spatial sketch was held constant while the text-based feature conditioning was iteratively modified. As illustrated in Fig. 4, the model dynamically generates entirely distinct facial characteristics—specifically adapting to different hair colors and textures—solely based on the provided text, proving that our architecture synthesizes features dynamically rather than relying on a static, biased dataset prior.

Fig. 4: From left to right: the original input sketch, followed by outputs dynamically conditioned on "dark black", "light brown", and "blonde" hair, respectively.

2) Impact of the Forensic Domain-Adaptation LoRA: To actively mitigate the inherent aesthetic bias of the foundational diffusion model, we integrated a custom-trained forensic LoRA adapter. To evaluate its efficacy, we conducted an ablation study on the adapter's influence by systematically scaling its integration weight (α). As demonstrated in Fig. 5, increasing the weight effectively suppresses the generative "beautification prior," introducing raw skin textures, asymmetric imperfections, and a more realistic phenotypic rendering. During this progression, all other textual and structural conditioning parameters—including supplementary adapters for skin tone and age—were held strictly constant. Through empirical evaluation, we determined that an optimal balance between spatial coherence and forensic realism is achieved at a LoRA weight of $\alpha = 0.7$.As the weight increases, the model successfully strips away the artificial aesthetic smoothness, resulting in a highly unretouched, forensically accurate output.

Fig. 5: Qualitative ablation of the forensic LoRA adapter weight. The top-left panel displays the original input forensic sketch, while the bottom-right panel displays the ground-truth target photograph. The six intermediate panels illustrate the progressive effect of the LoRA adapter as its weight (α) is scaled from 0.0 to 1.0 in increments of 0.2.

3) Skin Tone Adaptation: Beyond mitigating foundational aesthetic biases, true zero-shot forensic synthesis requires the ability to precisely isolate and manipulate specific demographic traits without degrading the underlying

biometric identity. To demonstrate this disentangled control, we evaluated the influence of our specialized skin-tone LoRA adapter. By systematically sweeping the adapter weight (α) across a continuous scale, we can dynamically interpolate the subject's pigmentation. As illustrated in Fig. 6, scaling the weight from $\alpha = -2.0$ to $\alpha = 3.0$ results in a monotonic shift from lighter to darker skin tones.

Fig. 6: Qualitative demonstration of disentangled skin-tone modulation. The top-left panel provides the input forensic sketch, and the bottom-right panel displays the ground-truth target photograph. The six intermediate panels (read left-to-right, top-to-bottom) illustrate the effect of scaling the skin-tone LoRA adapter weight (α) from -2.0 to 3.0 in increments of 1.0.

4) Temporal Adaptation and Age Modulation: In forensic investigations, particularly prolonged missing person inquiries or cold cases, the ability to accurately simulate the effects of chronological aging on a suspect is of paramount importance. To facilitate this temporal adaptation within our framework, we integrated a specialized age-modulation LoRA adapter. We conducted an ablation study by systematically scaling the adapter weight (α) to evaluate its capacity for disentangled age progression and regression. As depicted in Fig. 7, adjusting α dynamically alters age-correlated phenotypic markers—such as variations in skin elasticity, textural wrinkling, and localized structural sagging.

VII. Discussion

A. Strengths of the Current Pipeline

The current system demonstrates several important strengths:

- Effective sketch-to-image conditioning through lineart ControlNet.
- Good support for long and structured verbal descriptions.
- Stronger skin realism than generic SD 1.5 baselines.
- Practical modularity, allowing insertion of VAE, text encoder, ControlNet, and LoRAs.
- Straightforward side-by-side visualization for rapid qualitative analysis.

Fig. 7: Qualitative demonstration of temporal age modulation. The original input forensic sketch (top-left) and ground-truth target photograph (bottom-right) are displayed alongside intermediate generated outputs with varying age LoRA weights (α).

B. Current Limitations

Despite promising outputs, several limitations remain: identity faithfulness is not guaranteed; text compliance can still be inconsistent for rare or subtle attributes; outputs may overfit the realism prompt and occasionally become overly harsh; the system still depends on prompt engineering for suppressing beautification; and there is not yet a rigorous quantitative identity evaluation.

VIII. Future Work

Potential next steps include:

1. Reintroducing and systematically evaluating scheduler variants such as DPM-Solver SDE with Karras sigmas.
2. Comparing multiple ControlNet preprocessing methods and conditioning scales.
3. Improving caption normalization and ontology consistency for LoRA training.
4. Building a structured benchmark with human evaluation.
5. Incorporating identity-oriented constraints or embedding-based similarity evaluation.

IX. Conclusion

This project developed a modular forensic sketch-to-photo pipeline that combines a photorealistic Stable Diffusion 1.5-family model, an improved VAE, ControlNet lineart conditioning, and LongCLIP-based long-text understanding. The resulting system is capable of transforming line sketches and structured clinical descriptions into photorealistic facial images while explicitly suppressing beautification bias through prompt design.

In addition to the core inference pipeline, the project also established a practical path toward domain adaptation using a custom LoRA trained on mugshot-style images and clinical captions. This LoRA extension is intended to improve forensic realism and style specialization without requiring full-model fine-tuning.

Overall, the project demonstrates the feasibility of a controllable AI-based forensic sketch reconstruction workflow and provides a strong basis for future refinement in identity preservation, robust evaluation, and deployment.

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Appendix

Throughout the development of the proposed forensic facial synthesis pipeline, several alternative architectures, training strategies, and baseline models were evaluated. Many of these approaches were ultimately discarded due to poor empirical performance, largely constrained by dataset size and domain-specific challenges. This section details these exploratory attempts and the rationale for their exclusion.

A. Benchmarking Existing Frameworks

We initially evaluated existing sketch-to-image frameworks, specifically DeepFace Drawing and DiffFaceSketch. However, empirical testing revealed that these models are fundamentally optimized for sparse, abstract scribbles rather than highly detailed forensic line-art. When applied to intricate police sketches, the models exhibited structural confusion, resulting in outputs that failed to retain the necessary biometric geometry.

B. Coarse-to-Fine Generation Attempts

Inspired by recent literature, we experimented with a two-stage coarse-to-fine generation pipeline. A CycleGAN was trained to provide an initial colorized prior, which was then passed to a Denoising Diffusion Probabilistic Model (DDPM) for refinement. This approach suffered from a severe "averaging effect" during the coarse stage. The CycleGAN output provided weak, generalized guidance to the diffusion model, ultimately leading to generative hallucinations and a complete loss of identity fidelity.

C. Direct Diffusion and Sketch Encoder Training

Attempts were made to train a custom sketch encoder from scratch with the goal of aligning sketch embeddings directly into a photorealistic embedding space (utilizing InsightFace Buffalo and iResNet backbones). We evaluated several encoder architectures, including ResNet50, VGGFace2, and iResNet. While iResNet yielded the most promising initial spatial alignments, all custom architectures ultimately suffered from severe overfitting due to the limited size of our paired sketch-photo dataset ($\sim$2,600 samples). Similarly, training a custom DDPM pipeline natively on this small dataset proved ineffective for learning robust facial priors.

D. ControlNet Modalities and Scheduler Selection

During the development of the conditioning pipeline, we tested various ControlNet modalities, including Scribble, Canny, and standard Lineart. Early iterations utilizing standard Stable Diffusion base models produced outputs that were overly artistic, stylized, and lacked forensic realism. To mitigate this aesthetic bias, we transitioned the base model to `Realistic_Vision_V5.1`.

Furthermore, we conducted an ablation study on diffusion inference schedulers to improve skin texture and fine details. While the `UniPCMultistepScheduler` offered fast convergence, the `DPMSolverSDEScheduler` was ultimately selected for the final pipeline due to its superior ability to synthesize high-frequency details, such as realistic skin texture and pores, without introducing artifacts.